GENX: Porewater Ammonium

May 2026 Samples

2026-07-23

Contents

##Run Information

```r
cat("Run Information: Brenden Blakley") #lets you know what section you're in
```

```
## Run Information: Brenden Blakley
```

```r
library(here)
```

```
## Warning: package 'here' was built under R version 4.5.3
```

```
## here() starts at C:/Users/blakleybd/Documents/GitHub/GCREW
```

```r
#set the run date & user name
  sample_year <- "2026"
  sample_month <- "May"
  user <- "Brenden Blakley"

#identify the files you want to read in
  #read in as a list to accommodate multiple runs in a month
  NH3_files <- c("Raw Data/20260608_2m_GENXrr_May_NH4.csv",
                 "Raw Data/20260624_GENXrr_2Mrr_May2026.csv")

# Define the file path for QAQC log file - NO Need to change just check year
  file_path <- here("Porewater QAQC Logs", "GCReW_NH4_QAQC_Log.csv")
  file_path_QAQC <- here("Porewater QAQC Logs", "GCReW_SEAL_Check_QAQC_Log.csv")
  final_path <- "Processed Data/GCReW_GENX_Porewater_NH4_202605.csv"

#record any notes about the run or anything other info here:
  run_notes <- ""

  #Collection_Dates = "Raw Data/Sample_Collection_Dates_2024.csv"

  cat(run_notes)
```

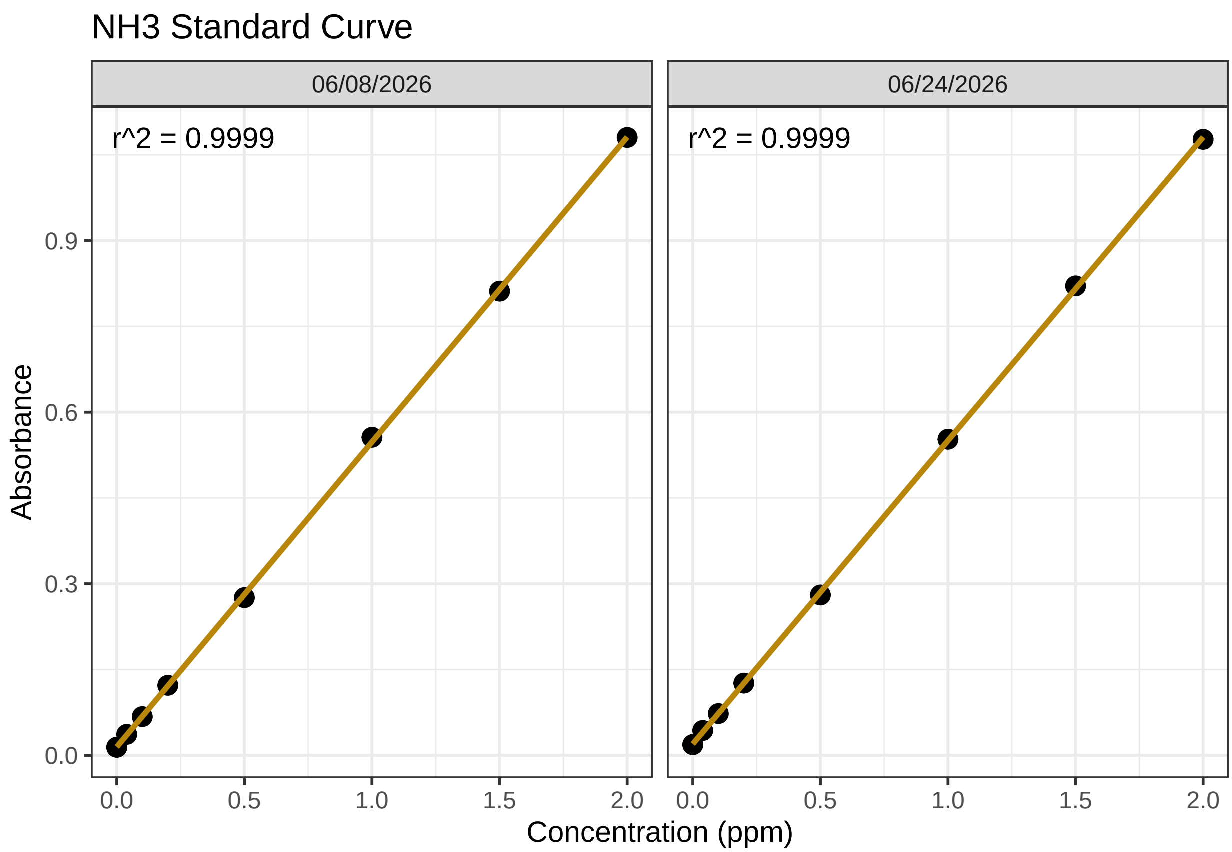
##Setup

##Read in metadata and create similar sample IDs for matching to samples

##Import Data & Clean

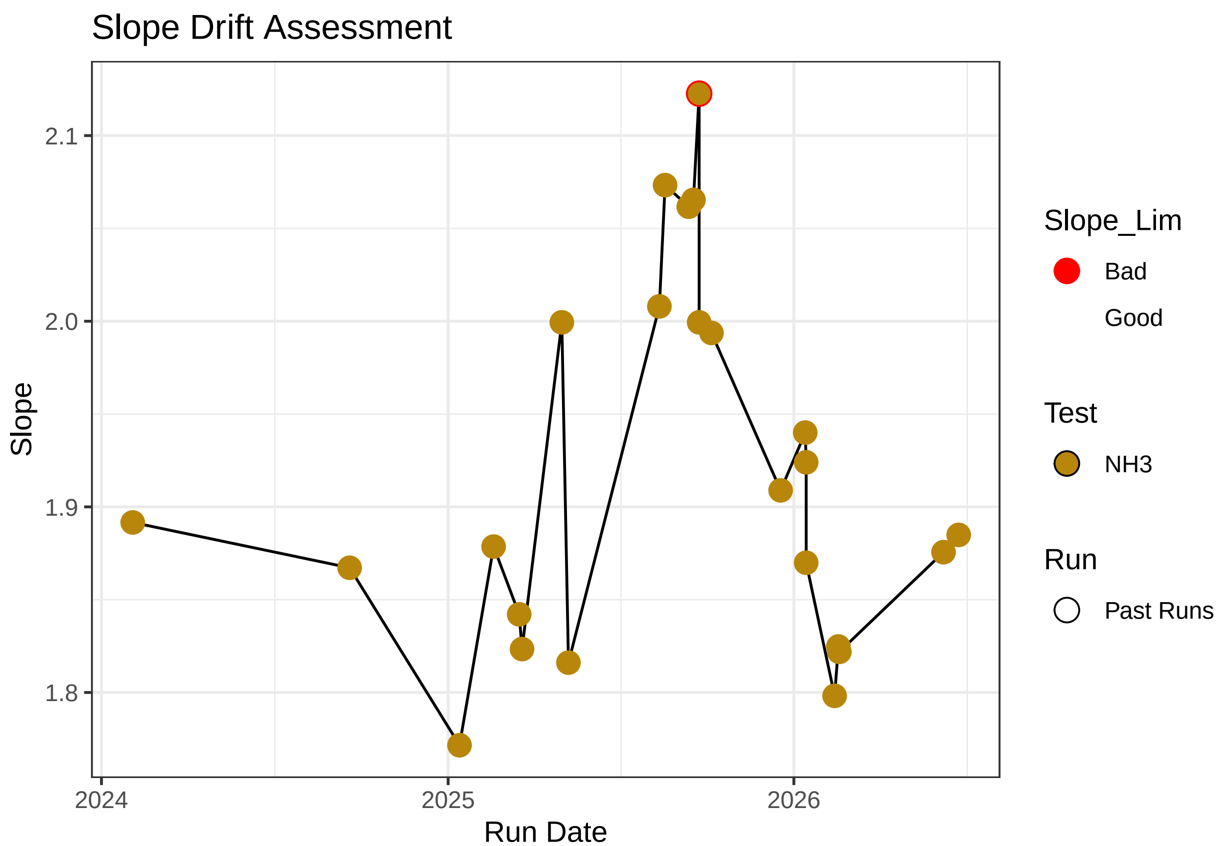
0.1 Assessing standard Curves

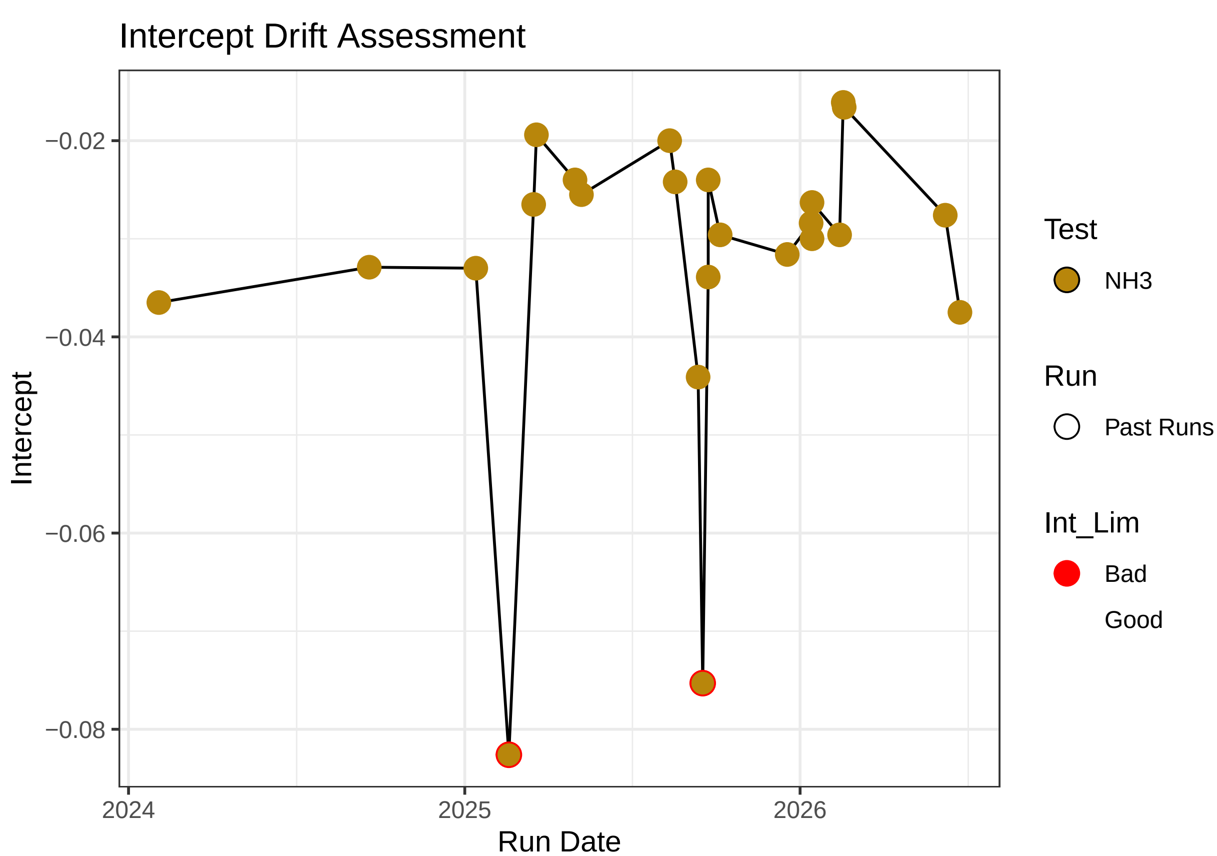
```
## 'geom_smooth()' using formula = 'y ~ x'
```



Test	Run_Date	R2_Flag	R2	Slope
Ammonia 2	06/08/2026	R2_Pass - PROCEED	0.9998997	1.875603
Ammonia 2	06/24/2026	R2_Pass - PROCEED	0.9999260	1.884907

0.2 Update Standard Log





0.3 Dilution Corrections - ensure the latest dilution is kept

```
## [1] "Reran Samples: c(\"Ch4-80\", \"Ch5-80\", \"Ch6-40\", \"Ch6-80\", \"Ch8-40\", \"Ch8-80\", \"Ch9-8
```

```
## [1] "Diluted Samples: c(\"Ch4-80\", \"Ch5-80\", \"Ch6-40\", \"Ch6-80\", \"Ch8-40\", \"Ch8-80\", \"Ch9
```

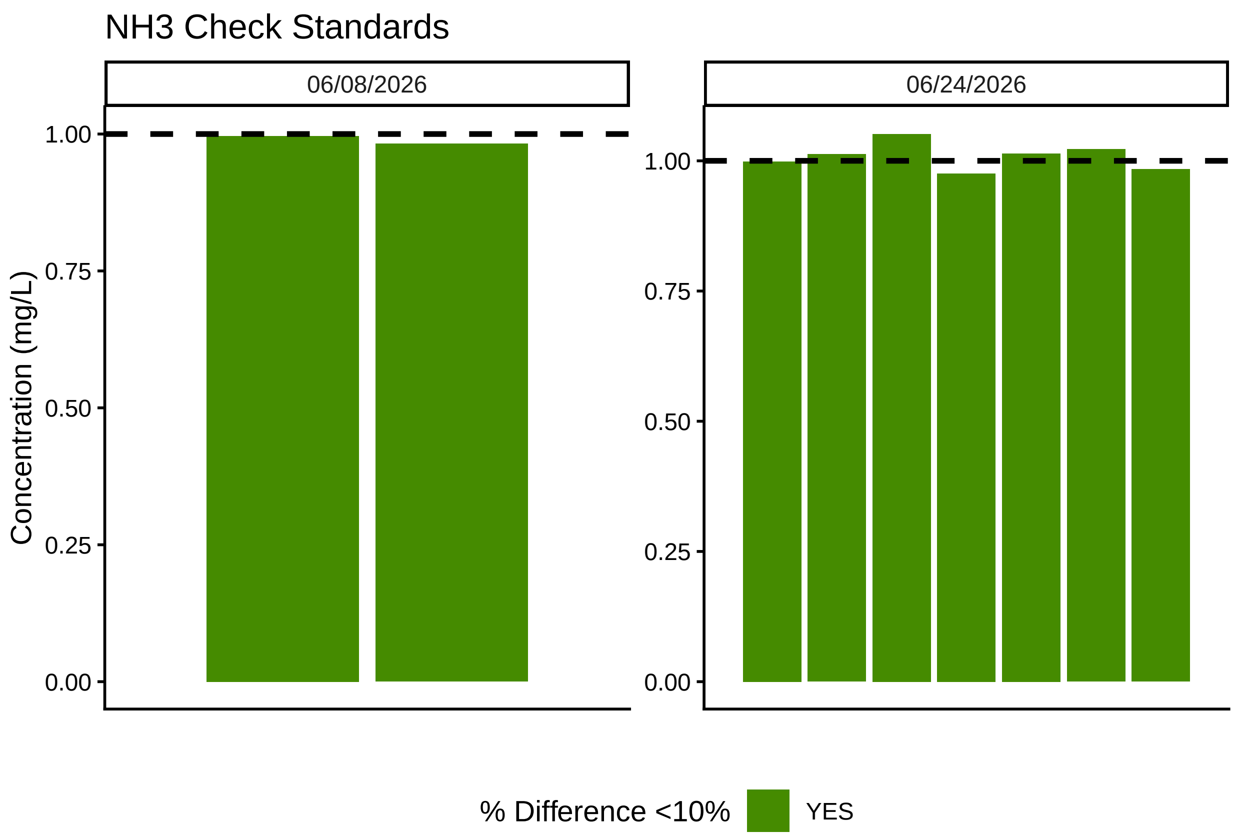
```
## [1] "No Naming Issues Detected"
```

0.4 Performance Check

Test	Run_Date	PE_Flag	PE_Conc	PE_Target_Conc
Ammonia 2	06/24/2026	Performance Check Within 25% - PROCEED	1.148001	1.034

0.5 Analyze the Check Standards

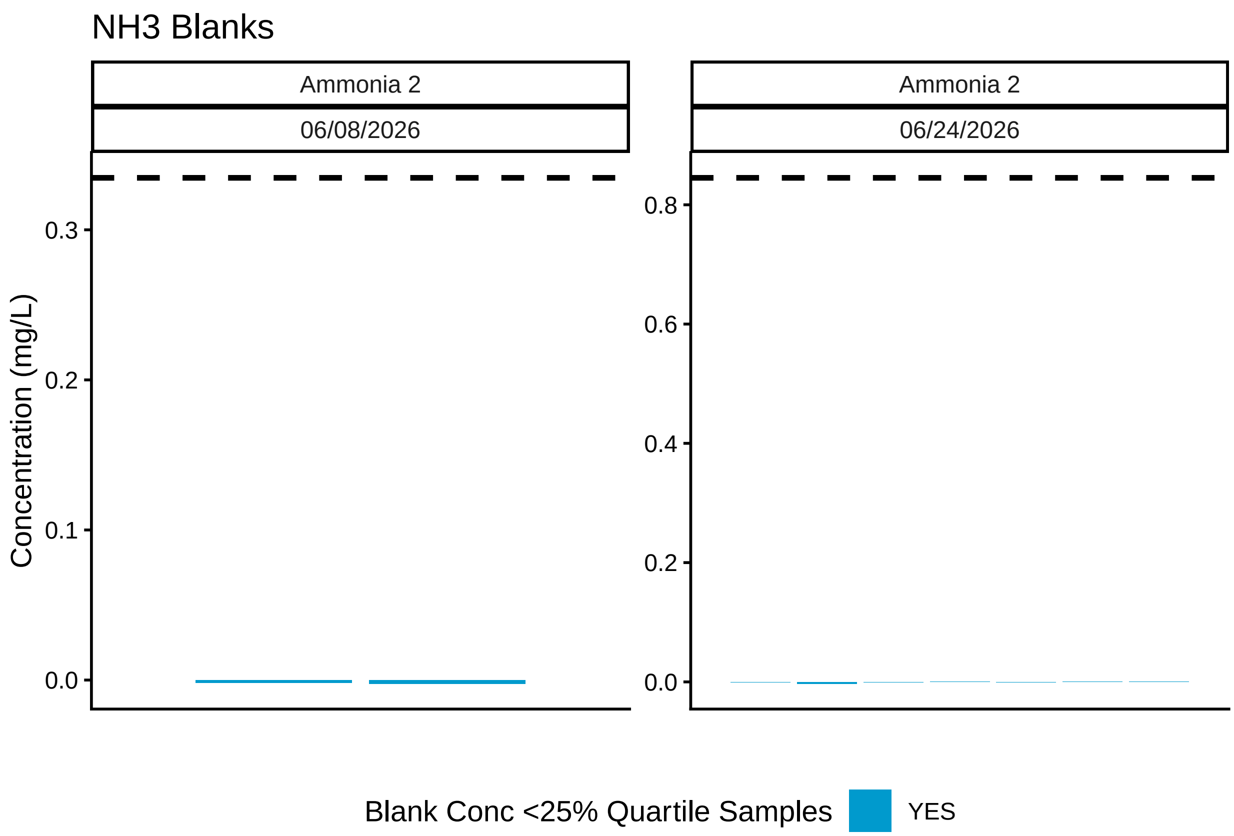
Test	Run_Date	RSV_Flag	RSV	RSV_Cutoff
Ammonia 2	06/08/2026	RSV WITHIN RANGE - PROCEED	0.0236851	0.25
Ammonia 2	06/24/2026	RSV WITHIN RANGE - PROCEED	0.0236851	0.25



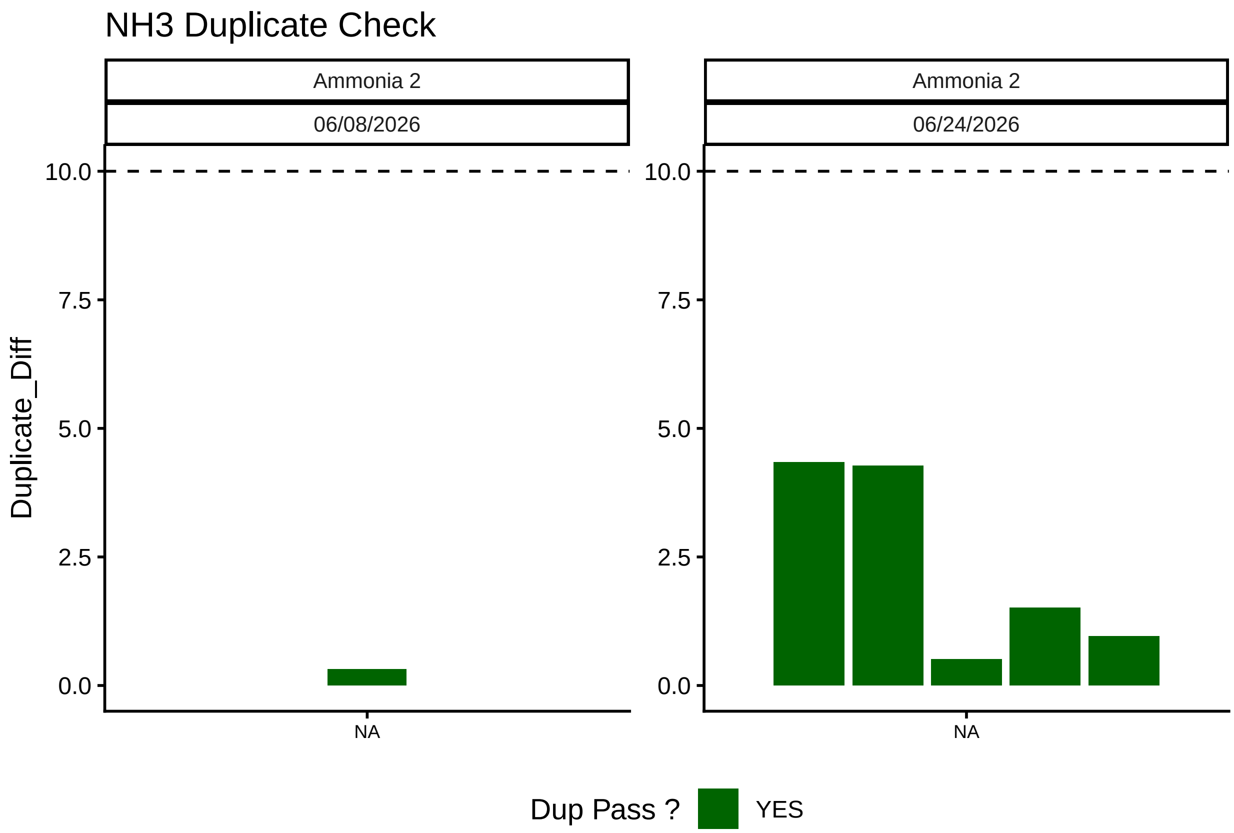
Test	Run_Date	CHK_Flag
Ammonia 2	06/08/2026	>60% of Checks Pass - PROCEED
Ammonia 2	06/24/2026	>60% of Checks Pass - PROCEED

0.6 Analyze Blanks

Test	Run_Date	BLK_Pct_Flag	Mean_Blk_Conc	Quantile_25
Ammonia 2	06/08/2026	>60% of Blanks Pass - PROCEED	-0.002067	0.3346265
Ammonia 2	06/24/2026	>60% of Blanks Pass - PROCEED	-0.000556	0.8452105

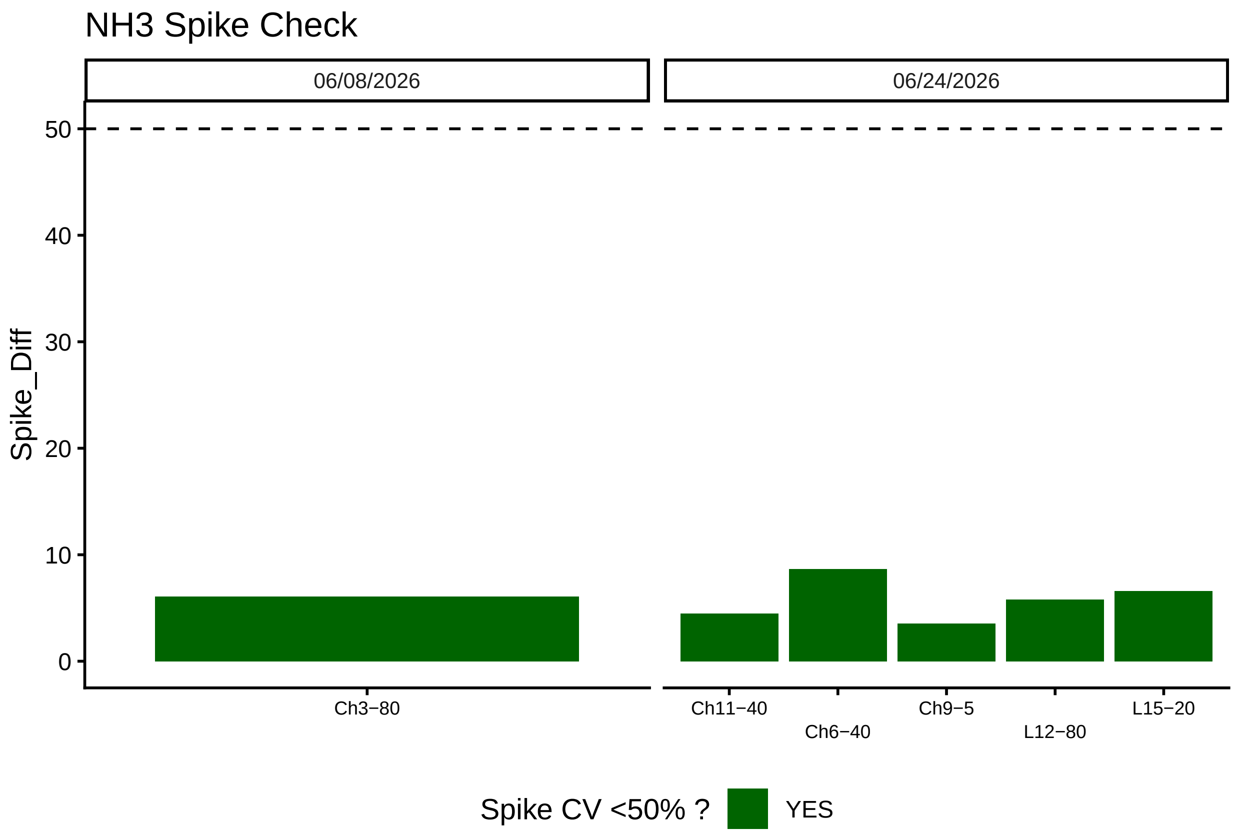


0.7 Analyze Duplicates



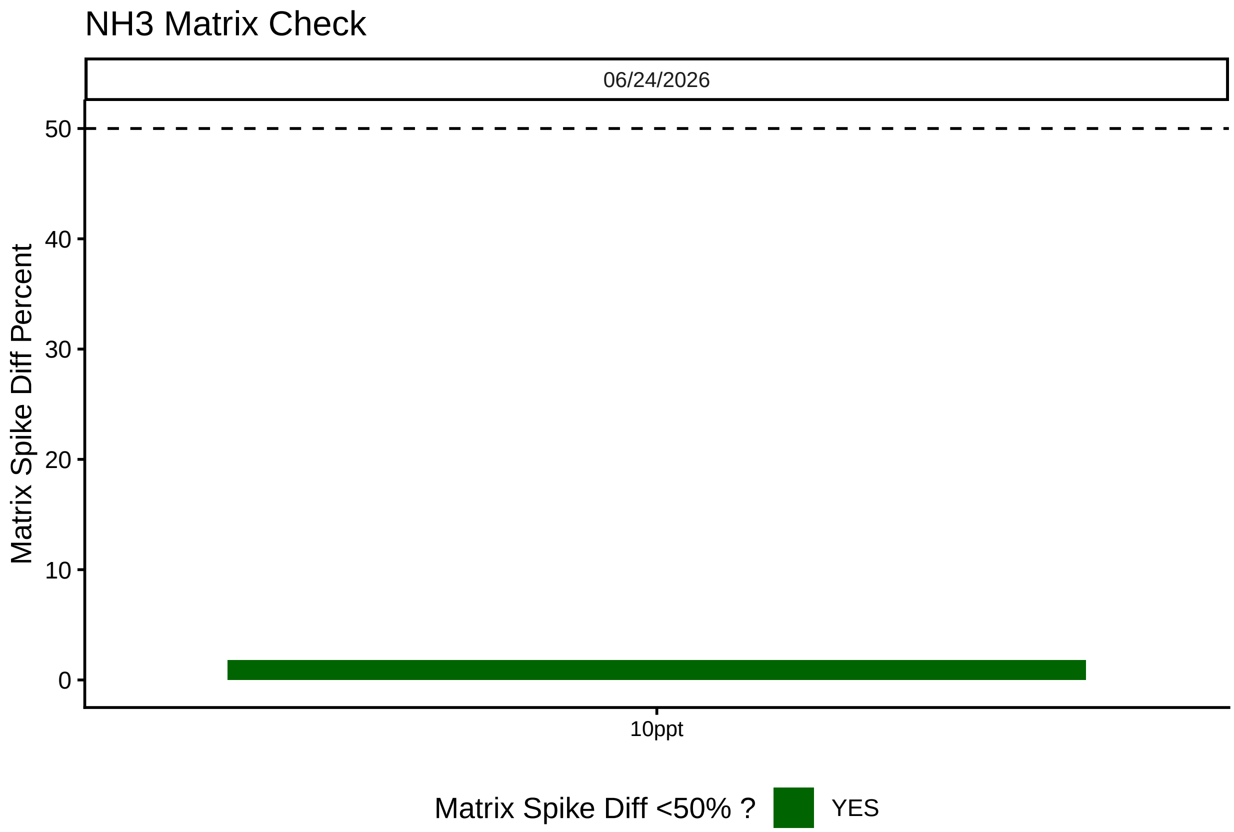
Test	Run_Date	Dup_Flags
Ammonia 2	06/08/2026	>60% of Dups Pass - PROCEED
Ammonia 2	06/24/2026	>60% of Dups Pass - PROCEED

0.8 Spikes



Test	Run_Date	Spike_Flags
Ammonia 2	06/08/2026	>60% of Spikes have a Diff <50% - PROCEED
Ammonia 2	06/24/2026	>60% of Spikes have a Diff <50% - PROCEED

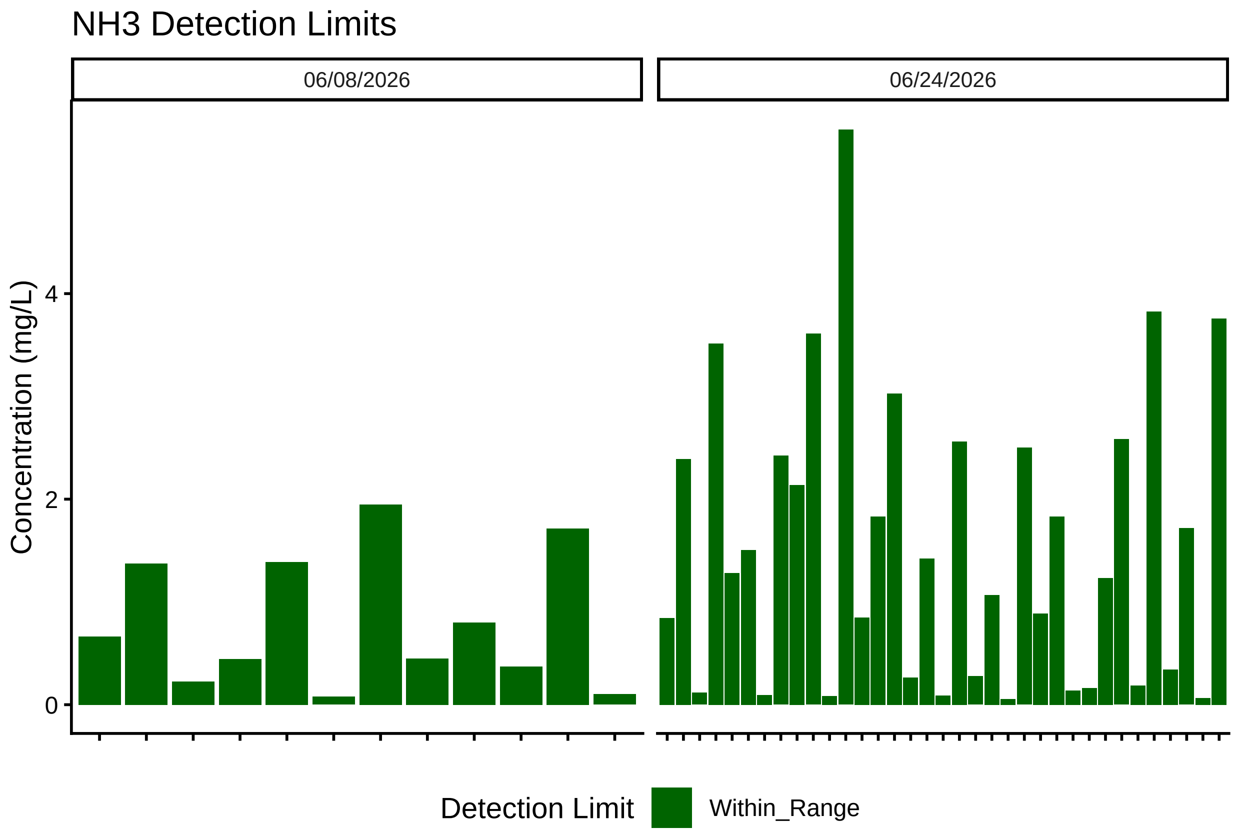
0.9 Matrix Effects



Test	Run_Date	Matrix_Flags
Ammonia 2	06/24/2026	Matrix Has Diff <50% - PROCEED

##Unit Conversion

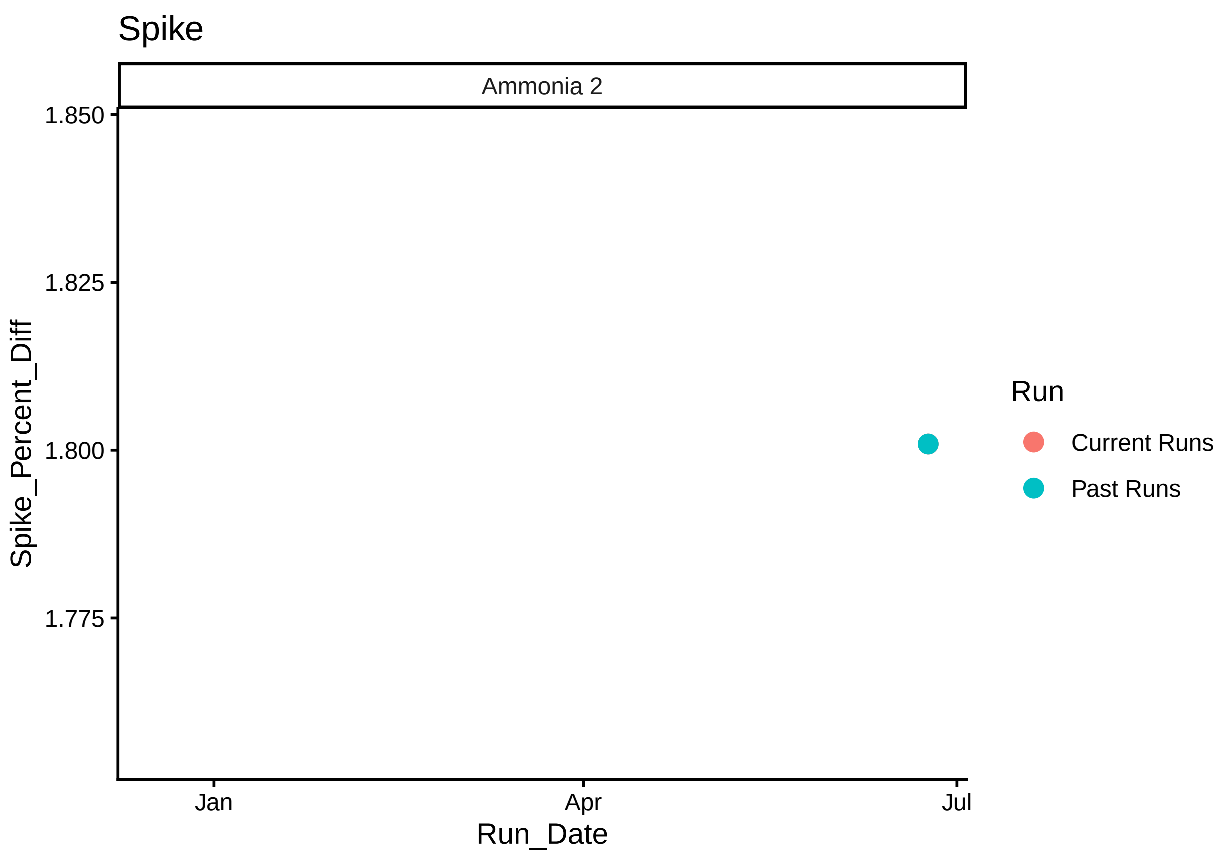
##Sample Flagging - Within range of standard curve

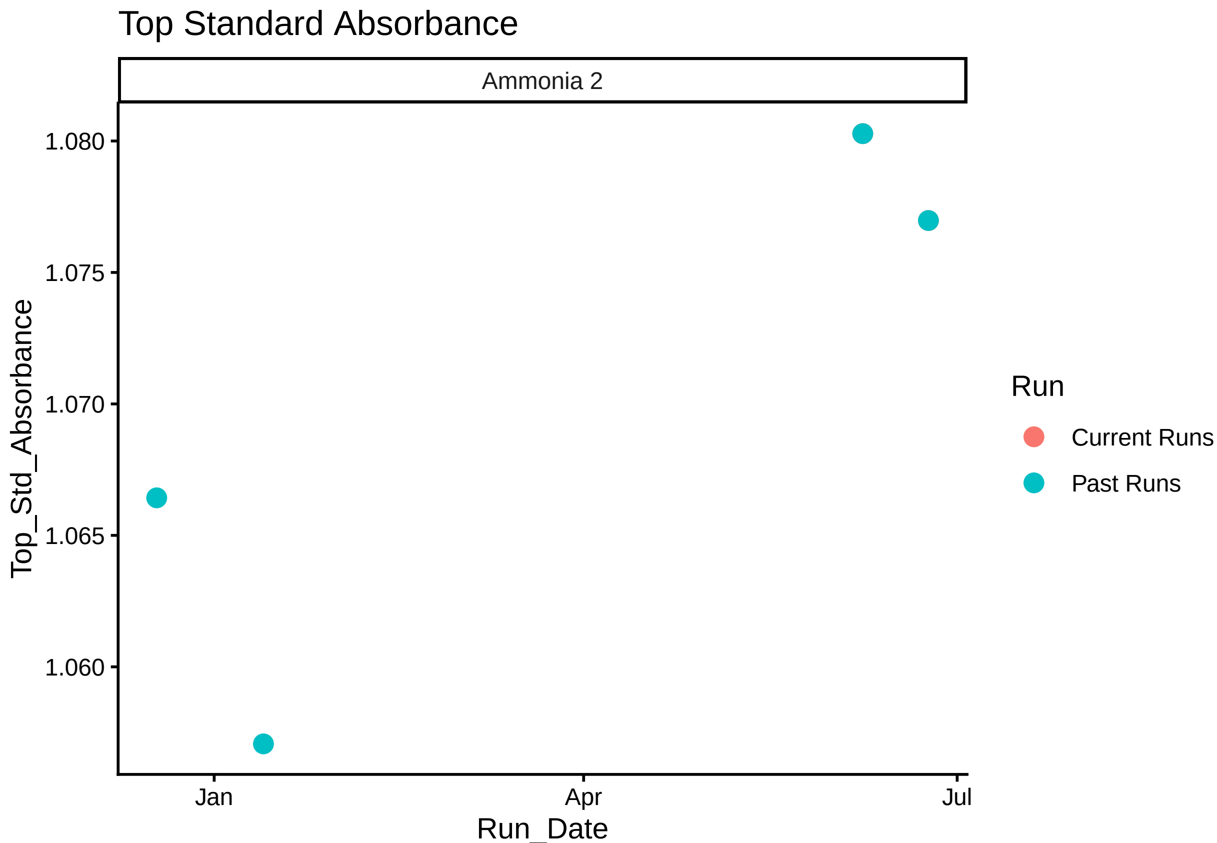


##Add QAQC Flags

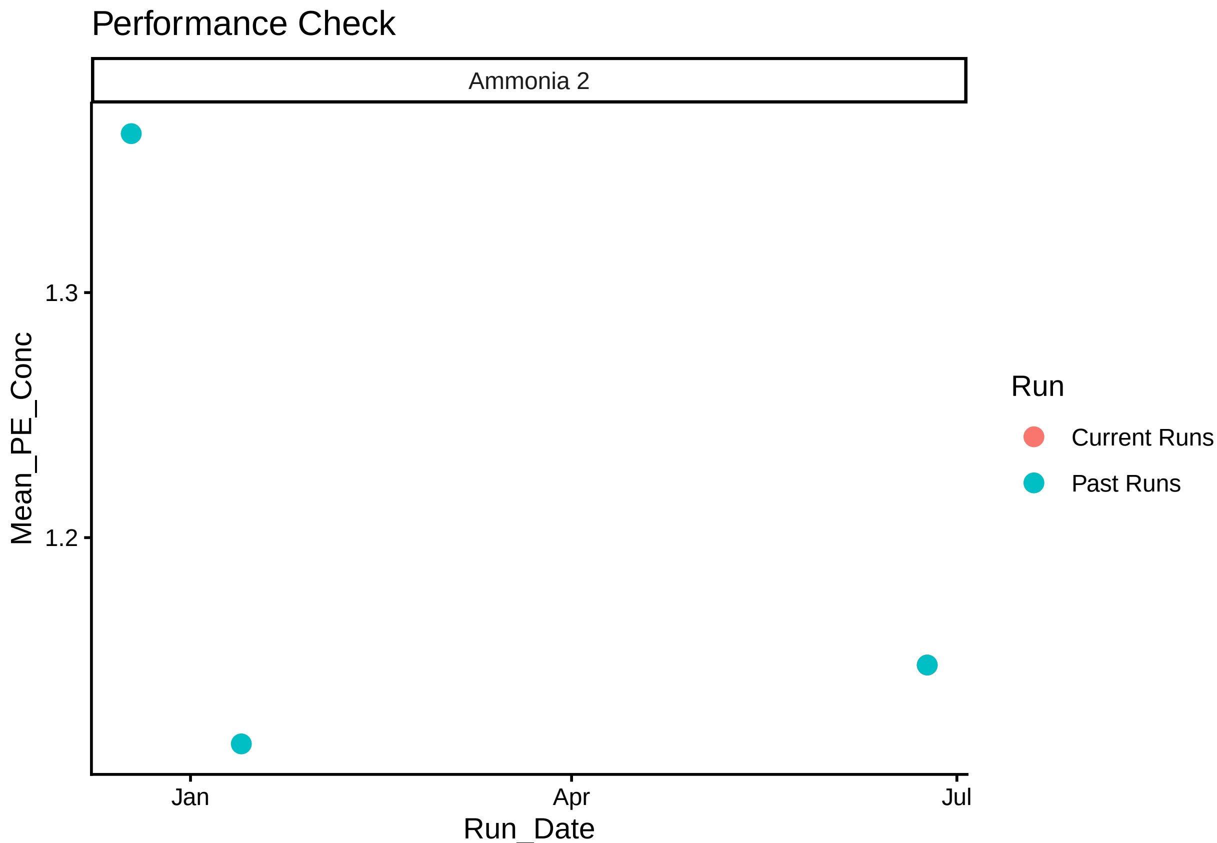
0.10 Create Log for QAQC

```
## Warning: Removed 4 rows containing missing values or values outside the scale range
## (‘geom_point()‘).
```


Spike
Ammonia 2
1.850
1.825
1.800
1.775
Spike_Percent_Diff
Jan
Apr
Jul
Run_Date
Run
Current Runs
Past Runs




```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



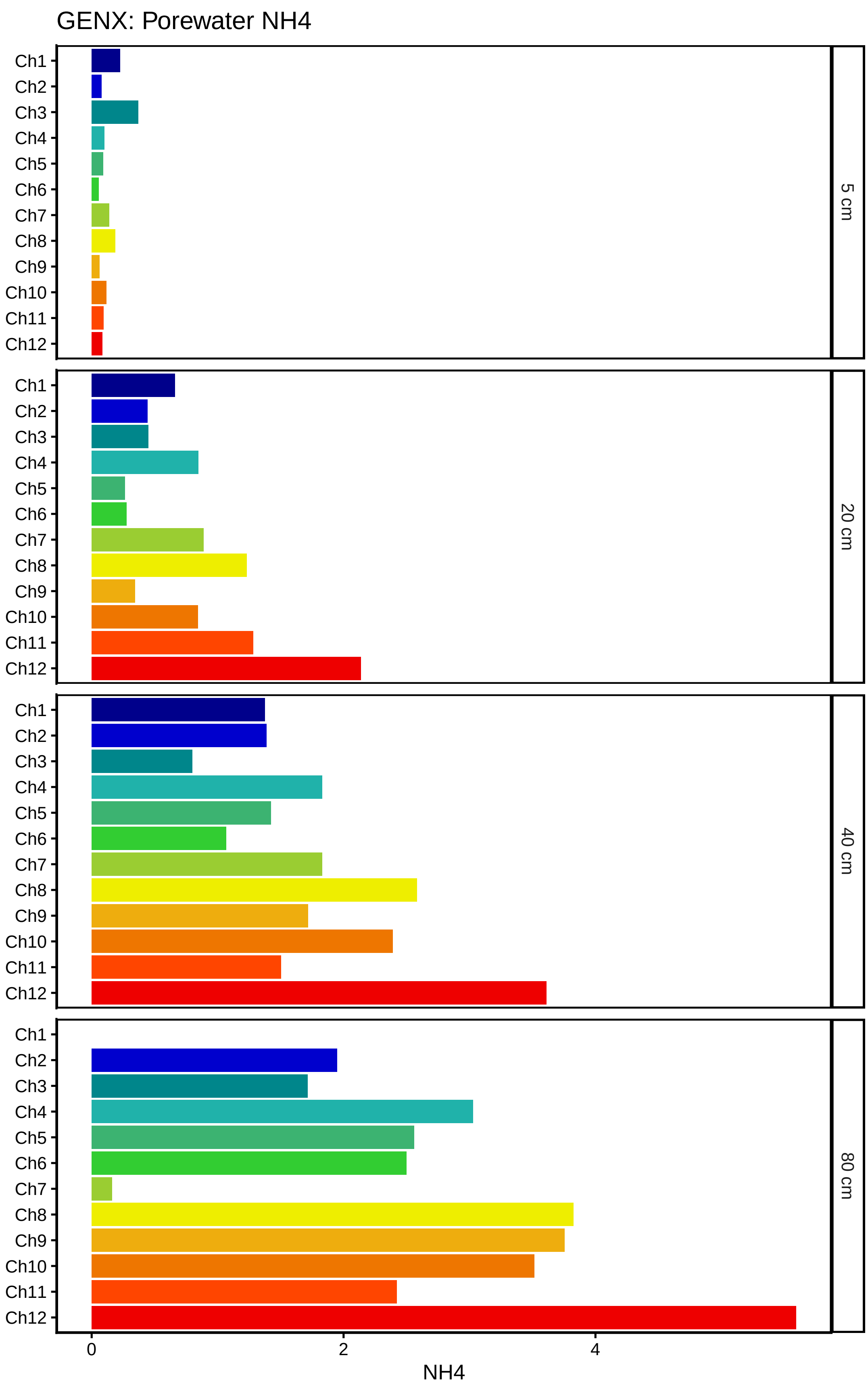
0.11 Merge samples with Metadata & check all samples are present

```
## Merge Samples with Metadata

## Some sample IDs are missing from data.

## [1] "Ch1-80"
```

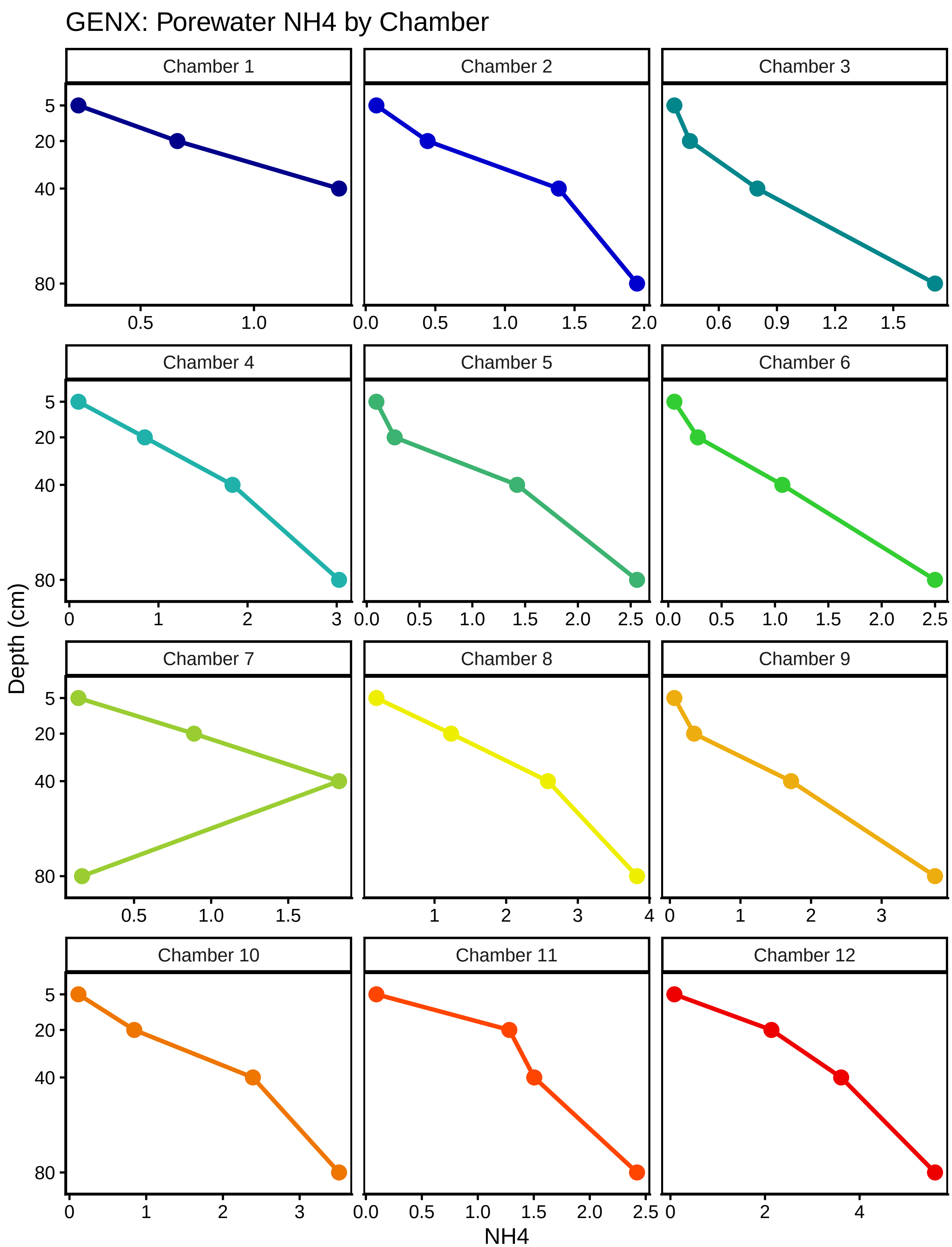
0.12 Visualize Data

```
## Visualize Data
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (‘geom_col()‘).
```


GENX: Porewater NH4
Ch1
Ch2
Ch3
Ch4
Ch5
Ch6
Ch7
Ch8
Ch9
Ch10
Ch11
Ch12
5 cm
20 cm
40 cm
80 cm
0
2
4
NH4

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (‘geom_point()‘).

## Warning: Removed 1 row containing missing values or values outside the scale range
## (‘geom_path()‘).
```


GENX: Porewater NH4 by Chamber
Chamber 1
Chamber 2
Chamber 3
Chamber 4
Chamber 5
Chamber 6
Chamber 7
Chamber 8
Chamber 9
Chamber 10
Chamber 11
Chamber 12
Depth (cm)
5
20
40
80
NH4

0.13 Organize

#Export Data

#end